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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)
A.Y. 2026 – 2027

TITLE OF THE PROJECT	**Smart Campus: Placement Management System**	
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ABSTRACT

Campus placement is an important process in educational institutions that involves students, recruiters, placement coordinators, and administrators. Managing student profiles, academic information, skills, company details, job opportunities, eligibility criteria, applications, interview schedules, and selection results through manual or disconnected systems can be time-consuming and difficult to maintain. Students may face difficulties in finding suitable job opportunities and tracking their application status, while recruiters and placement coordinators need an efficient way to manage large numbers of students and recruitment activities. These challenges highlight the need for a centralized and efficient placement management system that can simplify the overall campus recruitment process.

The proposed Smart Campus: Placement Management System is a service-oriented, microservices-based application designed to provide a centralized platform for managing campus placement activities. The system enables students to create and maintain their profiles by providing details such as academic qualifications, skills, certifications, and other relevant information. Students can view available job opportunities, check their eligibility based on company requirements, apply for suitable positions, and track the status of their applications throughout the recruitment process. The system also enables recruiters to manage company information, create job postings, specify eligibility requirements, view applications, shortlist candidates, and update recruitment outcomes.

The system is designed using multiple independent services to separate different business functionalities and improve maintainability and scalability. The major services include Student Service, Company and Job Service, Application Service, and Notification Service. The Student Service manages student profiles and academic information, while the Company and Job Service manages recruiter and company details along with available job opportunities. The Application Service handles student applications, shortlisting, interview status, and selection information.

The Notification Service provides updates related to applications, interviews, shortlisting, and selection. These services communicate through REST APIs and work together to provide a complete placement management workflow.

The application uses Java, Spring Boot, and Spring Cloud for developing the backend microservices. Eureka Service Discovery is used for registering and discovering services, while Spring Cloud Gateway provides centralized routing for client requests. Spring Security and JWT-based authentication are incorporated to provide secure user authentication and role-based authorization for students, recruiters, and administrators. PostgreSQL is used for storing application data, following a database-per-service approach to maintain separation between individual microservices. The system also incorporates REST API testing and automated testing practices to ensure the reliability and correctness of the services.

The proposed system provides a structured and efficient approach to campus placement management by reducing manual effort, improving accessibility of placement information, and enabling users to track recruitment activities in a centralized platform. The microservices architecture allows individual services to be developed, maintained, tested, and scaled independently. By integrating secure authentication, service discovery, API gateway routing, independent databases, and REST-based communication, the Smart Campus: Placement Management System aims to provide a reliable, scalable, and user-friendly solution for managing the complete campus placement process.

Project Architecture/Project Overview: